

Oxalate Synthesis and Pyrolysis: A Colorful Introduction to Stoichiometry

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Facility in stoichiometry, that is, the ability to interconvert between amounts of chemicals, is an essential skill that is taught within the general chemistry curriculum and lays the foundation on which advanced chemistry knowledge is built (1). As outlined in Taft's 1993 National Curriculum Survey, a majority of U.S. undergraduate institutions moderately or extensively cover stoichiometric calculations in general chemistry courses (2). Consequently, coordinated stoichiometry-based laboratory experiments have been developed and implemented into the general chemistry curriculum for over five decades (3–7) with development continuing to the present (8–13). However, large enrollment general chemistry classes necessitate the use of diverse themes to appeal to a wide variety of student interests. Thus, contextual examples highlighting the importance and usefulness of the chemicals involved in stoichiometry experiments lend relevance to the discussion and enhance student interest. For example, Bracken and Tietz developed a stoichiometry experiment in which students determined the mass percent of $2\text{Na}_2\text{CO}_3/3\text{H}_2\text{O}_2$ (active ingredient) in commercial stain removers (8). Use of “chemicals in the home” lent relevance to the experiment and enhanced student interest.

Metal oxalates represent a chemical system that allows students to contextualize chemistry within their everyday lives (Table 1). From kidney stones (14–18) and metal oxide nanoparticle formation (19–21) to oxalate's suspected role in joint inflammation (22) and autism (23, 24) and to their use as pigments and as support for life on Mars (25), the importance of oxalates is evident. Oxalate compounds are appropriate for use within the general chemistry laboratory because of their straightforward connections to society.

Metal oxalate synthesis is well documented within the literature (28–31) and the route from metal oxalates to metal oxides is often accomplished by pyrolysis, that is, causing a chemical reaction to occur by heat (32). The pyrolysis of transition-metal oxalates is relatively simple, and many oxalates are known to give well-defined oxide products (33–36).

We report here on an original stoichiometry experiment that is based upon the synthesis and subsequent pyrolysis of three metal oxalate compounds, iron(II) oxalate dihydrate, nickel(II) oxalate dihydrate, and manganese(II) oxalate trihydrate. Our objective was to design an experiment that would (i) assist students in learning stoichiometry, (ii) highlight the importance of proper stoichiometric calculations and

experimental techniques, and (iii) provide relevance to enhance student interest. In this experiment, metal oxalate synthesis and pyrolysis is followed by determination of the pyrolysis reaction and the identity of the pyrolysis product using stoichiometric calculations and comparisons of theoretical and actual yields.

Experimental Description

Prelaboratory Exercises

Each student is assigned one of three metal salts as the starting material for their oxalate synthesis: iron(II) chloride tetrahydrate [$\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$], nickel(II) nitrate hexahydrate [$\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$], or manganese(II) sulfate monohydrate [$\text{MnSO}_4 \cdot \text{H}_2\text{O}$]. Students complete a set of prelaboratory exercises specifically tailored to the stoichiometry of their assigned starting material. Students balance their salt's oxalate synthesis reaction (Table 2) and three potential pyrolysis reactions (Table 3) for their metal oxalate. Finally, students calculate theoretical product yields, based upon pyrolysis of 1.0 g of hydrated metal oxalate, for each pyrolysis reaction. These theoretical yields are referenced later in the experiment.

Synthesis of the Metal Oxalate Compound

Students dissolve approximately 4.0 g of their assigned metal salt into 20.0 mL of distilled water. This solution is slowly poured, with stirring, into 50.0 mL of 0.888 M oxalic acid aqueous solution to precipitate the metal oxalate. Vacuum filtration follows with quantitative transfer of the metal oxalate onto a piece of preweighed filter paper within the Buchner funnel. Washing with distilled water is followed by several acetone rinses to expedite drying of the precipitate. Once the precipitate is completely dry, the filter paper and precipitate are weighed together and the actual yield of metal oxalate is obtained. Calculations of limiting reactant and theoretical and percent yields of metal oxalate ensue.

Pyrolysis of the Metal Oxalate Compound

Into a weighed aluminum dish, students place 0.99–1.01 g of the dry metal oxalate. This dish is placed on a hot plate and heated for a minimum of 10 min at ca. 275 °C to effect dehydration and begin pyrolysis. Higher temperature heating

Table 1. Oxalates in Context

Metal Oxalate	Effect or Use
Calcium oxalate	Kidney stones in humans and animals (14–18)
Transition-metal oxalates	As precursors to metal oxide nanoparticles (19–21): applications in catalysis (26) and semiconductors (27)
Iron(III) oxalate	Used as yellow pigment: paints, glass, and plastic
Naturally occurring metal oxalates, e.g., FeC_2O_4 (humboldtine)	Product of fungi/lichen growth: presence used as evidence for life on other planets (25)
Metal oxalate crystals	Formation in bones/joints: associated with inflammation (22), anemia, immunosuppression
Low oxalate diet	As treatment for autism: preliminary research indicates that autistic children have high amounts of oxalates in urine (23, 24)

Table 2. Balanced Oxalate Synthesis Reactions

Metal Salt	Metal Oxalate Synthesis Reaction
$\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$	$\text{FeCl}_2 \cdot 4\text{H}_2\text{O}(\text{aq}) + \text{H}_2\text{C}_2\text{O}_4(\text{aq}) \rightarrow \text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) + 2\text{HCl}(\text{aq})$
$\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	$\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}(\text{aq}) + \text{H}_2\text{C}_2\text{O}_4(\text{aq}) \rightarrow \text{NiC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}(\text{s}) + 2\text{HNO}_3(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$
$\text{MnSO}_4 \cdot \text{H}_2\text{O}$	$\text{MnSO}_4 \cdot \text{H}_2\text{O}(\text{aq}) + \text{H}_2\text{C}_2\text{O}_4(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{MnC}_2\text{O}_4 \cdot 3\text{H}_2\text{O}(\text{s}) + \text{H}_2\text{SO}_4(\text{aq})$

at ca. 400 °C for another 10 min follows, with subsequent cooling and reweighing. This procedure of heating, cooling, and reweighing is repeated until the mass of dish plus contents changes by no more than 10 mg for two successive cycles, that is, until constant weight is achieved. Because oxalate pyrolysis reactions occur with near 100% yield (28), the actual yield of pyrolysis product compares favorably with the theoretical yield for one of the pyrolysis products (Table 3) considered in the prelaboratory assignment. By comparison of actual and theoretical product yields, students identify the pyrolysis product and the true pyrolysis reaction.

Hazards

Goggles, apron, and gloves should be worn throughout this experiment. Students should be cautioned not to touch hot objects and should immediately place burns under cool running water. Oxalic acid may be fatal if swallowed, may cause kidney damage, and is corrosive and causes severe irritation and burns to the skin, eyes, and respiratory tract. Iron(II) chloride tetrahydrate and manganese(II) sulfate monohydrate are extremely destructive to tissues of the mucous membranes and upper respiratory tract in addition to being corrosive to the skin and eyes. Nickel(II) nitrate hexahydrate is a strong oxidizer and contact with other material may cause a fire. It is harmful if swallowed or inhaled, causes irritation to skin, eyes, respiratory tract, and is considered a carcinogen. Acetone is extremely flammable and causes irritation to skin, eyes, and respiratory tract. Students should thoroughly wash hands before leaving the laboratory.

Results and Discussion

Metal Oxalates and Pyrolysis Products

We undertook an extensive literature review to validate chemical identities for the metal oxalates (28–31) and oxalate pyrolysis products (33–36). In addition, synthesized metal

oxalates were analyzed via thermogravimetric analysis (heating rate: 10 °C min⁻¹). This analysis verified chemical formulas for iron and nickel oxalates as $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ and $\text{NiC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ and their pyrolysis products as Fe_2O_3 and NiO. However, this analysis indicated that the manganese oxalate was present in the trihydrate form, as $\text{MnC}_2\text{O}_4 \cdot 3\text{H}_2\text{O}$, instead of the expected dihydrate. Furthermore, the manganese oxalate's pink coloration and low dehydration temperature (<100 °C) confirmed the trihydrate (36). Research by Donkova and Mehandjiev into the thermal behavior of $\text{MnC}_2\text{O}_4 \cdot 3\text{H}_2\text{O}$ indicates that a mixture of Mn_3O_4 (major product) and Mn_2O_3 (minor product) is obtained upon oxalate pyrolysis at 450 °C (36). Given that Mn_3O_4 and Mn_2O_3 are similar in color (brown to brown–black) and upon pyrolysis of 1.0 g $\text{MnC}_2\text{O}_4 \cdot 3\text{H}_2\text{O}$ theoretical yields of Mn_3O_4 and Mn_2O_3 are comparable, 0.39 and 0.40 g, respectively, we chose to overlook this complication and assume the $\text{MnC}_2\text{O}_4 \cdot 3\text{H}_2\text{O}$ pyrolysis product was solely Mn_3O_4 .

Synthesis of the Metal Oxalate Compound

This experiment was implemented into a multisection, multi-instructor first-term general chemistry laboratory ($N = 509$). This course is primarily populated by first-year students majoring in one of the science, technology, engineering, or mathematics (STEM) disciplines. Statistical analyses of student experimental results for synthesis of each metal oxalate are shown in Table 4. Average percent yield values for oxalate synthesis are desirably high at 88.67%, 82.14%, and 53.65% for iron, nickel, and manganese oxalate hydrates, respectively. However, student percent yield values can be misleading because excess water within the sample or on the filter paper produces greater than expected yields (29). The use of acetone during the drying process did minimize this problem (28). Alternatively, the intermediate metal oxalate precipitate and filter paper could be dried using microwave radiation (37) or a low temperature drying oven, although temperatures would have to be maintained below 100 °C to preserve the hydrated forms. Thermogravimetric analysis (Table 3) of the oxalates synthesized via these

Table 3. Possible Reactions and Theoretical Product Yields for Metal Oxalate Pyrolysis

Metal Oxalate	Temp. of Dehydration and Pyrolysis	Possible Oxalate Pyrolysis Reactions	Theoretical Yield of Solid Product ^a
FeC ₂ O ₄ ·2H ₂ O yellow	dehydration: 178 °C ^b and 200 °C ^c pyrolysis: 268 °C ^b and 244 °C ^c	2FeC ₂ O ₄ ·2H ₂ O(s) + O ₂ (g) → 2FeO(s) + 4H ₂ O(g) + 4 CO ₂ (g) 4FeC ₂ O ₄ ·2H ₂ O(s) + 3O ₂ (g) → 2Fe ₂ O ₃ (s) + 8H ₂ O(g) + 8CO ₂ (g) FeC ₂ O ₄ ·2H ₂ O(s) → FeCO ₃ (s) + 2H ₂ O(g) + CO(g)	0.40 g FeO 0.44 g Fe ₂ O ₃ 0.64 g FeCO ₃
NiC ₂ O ₄ ·2H ₂ O blue–green	dehydration: 217 °C ^b and 227 °C ^c pyrolysis: 384 °C ^b and 353 °C ^c	2NiC ₂ O ₄ ·2H ₂ O(s) + O ₂ (g) → 2NiO(s) + 4H ₂ O(g) + 4CO ₂ (g) 4NiC ₂ O ₄ ·2H ₂ O(s) + 3O ₂ (g) → 2Ni ₂ O ₃ (s) + 8H ₂ O(g) + 8CO ₂ (g) NiC ₂ O ₄ ·2H ₂ O(s) → NiCO ₃ (s) + 2H ₂ O(g) + CO(g)	0.41 g NiO 0.45 g Ni ₂ O ₃ 0.65 g NiCO ₃
MnC ₂ O ₄ ·3H ₂ O light pink	dehydration: 98 °C ^b pyrolysis: 386 °C ^b and 321 °C ^c	2MnC ₂ O ₄ ·3H ₂ O(s) + O ₂ (g) → 2MnO(s) + 6H ₂ O(g) + 4CO ₂ (g) 3MnC ₂ O ₄ ·3H ₂ O(s) + 2O ₂ (g) → Mn ₃ O ₄ (s) + 9H ₂ O(g) + 6CO ₂ (g) MnC ₂ O ₄ ·3H ₂ O(s) → MnCO ₃ (s) + 3H ₂ O(g) + CO(g)	0.36 g MnO 0.39 g Mn ₃ O ₄ 0.58 g MnCO ₃

^aTheoretical yields are based on pyrolysis of 1.0 g of hydrated metal oxalate. ^bDehydration and pyrolysis temperatures obtained from thermogravimetric analysis on oxalate samples (nonuniform particle sizes) synthesized using the procedures outlined in this manuscript. ^cValues obtained from ref 33.

Table 4. Student Experimental Results for Synthesis of Metal Oxalates

Metal Oxalate	Mean Percent Yield	Median Percent Yield	Standard Deviation (%)	Number of Report Sheets ^a
FeC ₂ O ₄ ·2H ₂ O yellow	88.67	88.80	6.40	N = 165 (18 outliers excluded)
NiC ₂ O ₄ ·2H ₂ O blue–green	82.14	83.45	12.98	N = 151 (15 outliers excluded)
MnC ₂ O ₄ ·3H ₂ O light pink	53.65	53.31	12.64	N = 151 (7 outliers excluded)

^aExperimental data from N = 509 report sheets were analyzed. Outliers were identified, using the 1.5 interquartile range method, and were excluded from the statistical analysis. Two data sheets did not report any data.

procedures gave dehydration temperatures of 178, 217, and 98 °C for the iron, nickel, and manganese oxalates, respectively. As a result, any type of preliminary drying of the metal oxalates must be carried out at temperatures well below those for dehydration.

Pyrolysis of the Metal Oxalate Compound

Students determined the pyrolysis product's identity by comparison of pyrolysis product actual yield to calculated theoretical yields of three potential pyrolysis products (Table 3). Student mean actual yield values (Table 5) for both the iron and manganese pyrolysis products were 0.44 and 0.41 g, respectively, which compared favorably with theoretical yield values of 0.44 and 0.39 g. In addition, 78% and 76% of students would have chosen the correct iron and manganese pyrolysis products, respectively, assuming correct stoichiometric and experimental calculations.

The teaching assistants and instructors noticed several common trouble spots in the pyrolysis. Namely students did not heat to constant weight and did not achieve high enough temperatures to achieve complete pyrolysis. This was particularly true in the case of nickel oxalate, which changed from its initial color of bluish-green through an intermediate yellow color and finally to black upon completion of pyrolysis. Of those students specifying colors for their pyrolysis product, 46 of 52 (88%) who chose the incorrect and heavier NiCO₃ product indicated a color other than black (e.g., yellow, dark green, etc.) for their pyrolysis product; whereas, 33 of 38 (87%) who chose the correct NiO product specified black as their pyrolysis product's color. Teaching assistants and instructors caught the systematic error early in the laboratory week and were able to instruct students to extend the nickel pyrolysis until the pyrolysis product turned black and constant mass was achieved.

Although there were at least 20 teaching assistants who assisted students in completing this experiment, only a limited number (N = 8) provided us with survey feedback on this experiment. Teaching assistant opinions about this experiment were generally positive with 62.5% (N = 8) indicating that this experiment was worthwhile for their students. All respondents (N = 6) indicated that this experiment should be used again as is or with minor revisions. We have since revised

Table 5. Student Experimental Results for Oxalate Pyrolysis

Characteristic	Metal Oxalate		
	$\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ N = 183	$\text{NiC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ N = 168	$\text{MnC}_2\text{O}_4 \cdot 3\text{H}_2\text{O}$ N = 158
Correct Pyrolysis Product and Color	Fe_2O_3 ; reddish-brown; "rust"	NiO ; black	Mn_3O_4 ; brownish-black
Theoretical Yield of Pyrolysis Product ^a	0.44 g	0.41 g	0.39 g
Mean Value for Student's Actual Yield of Pyrolysis Product	0.44 g N = 127; 56 data points excluded; 28 did not follow directions ^b and 28 outliers ^c	0.56 g N = 143; 25 data points excluded; 25 did not follow directions ^b (no outliers)	0.41 g N = 112; 46 data points excluded; 28 did not follow directions ^b and 18 outliers ^c
Standard Deviation in Student Actual Yield	0.01 g	0.17 g	0.03 g
Pyrolysis Products (M = Fe, Ni, or Mn)	Theorized Pyrolysis Product Based on Student Results (%) ^d		
MO	10 (N = 18)	42 (N = 70)	9 (N = 15)
M_2O_3 or M_3O_4	78 (N = 143)	5 (N = 9)	76 (N = 120)
MCO_3	12 (N = 22)	53 (N = 89)	15 (N = 23)

^aBased on correct pyrolysis equation and pyrolysis of 1.0 g of hydrated metal oxalate. ^bStudents who did not pyrolyze 0.99–1.01 g of metal oxalate were excluded because the theoretical analysis was based on 1.0 g. ^cOutliers were identified, using the 1.5 interquartile range method, and were excluded from the statistical analysis. ^dTheorized pyrolysis products, that is, the product each student should have chosen based upon correct stoichiometric calculations, were obtained by comparison of each student's actual yield (in g) of pyrolysis product versus the theoretical yields given in Table 3.

the experimental procedure to address common trouble spots and issues related to incomplete pyrolysis of the nickel oxalate dihydrate.

Conclusions

We report here on an original stoichiometry experiment that involves the synthesis and subsequent pyrolysis of transition-metal oxalates. One objective in developing this experiment was to assist students in learning stoichiometry. This objective was met by requiring students to (i) balance chemical reactions, (ii) find the limiting reactant, and (iii) calculate theoretical and percent yields of products. Fundamental stoichiometric skills, that is, converting between grams and moles using molar mass, and converting between chemicals using coefficients, were employed throughout this experiment.

A second objective was to highlight the importance of proper stoichiometric calculations and experimental technique. This objective was met when students' compared their pyrolysis products' theoretical yields (from stoichiometric calculations) to their actual yields (from the experiment). An error in either their stoichiometric calculation or their experimental technique resulted in noncorrespondence between theoretical and actual product yields and misidentification of the pyrolysis product.

Finally, we sought to provide relevance to enhance student interest. This experiment provides nice connections to society because of metal oxalate compounds' diversity of usages in everyday life. The experimental procedure is straightforward and utilizes common chemicals and equipment. Student results were fairly accurate in both the synthesis and pyrolysis of their metal oxalate compounds. Additionally, striking color changes (see Figure 1 in the supporting information) occur during the pyrolysis of the metal oxalate compounds, which are well received by students and serve to emphasize the chemical changes involved (38). Feedback solicited from the teaching assistants who oversaw this experiment was positive and all recommended that we continue using this experiment.

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Supporting Information Available

Experimental instructions for students; instructor notes; and grading key. This material is available via the Internet at <http://pubs.acs.org>.